

Project Objective:

The goal of this project is to build a secure, internal email infrastructure using **RHEL 8.10**. By integrating **Postfix** (MTA), **Dovecot** (MDA), and **SquirrelMail**, we have created a system where users can communicate via email through a browser-based interface without needing external internet-based mail services.

Why this project?

- **Centralized Communication:** To establish a private communication channel for an organization.
- **Security & Control:** To manage internal data without relying on third-party email providers.
- **Learning Integration:** To understand how SMTP, POP3, and IMAP protocols work together to deliver mail services.

Initially, I prepared the server environment to ensure all services could communicate without restriction.

1. **Hostname Setup:** Configured hostname as **nt67.edu**.
2. **IP Configuration:** Assigned Static IP **192.168.3.61**.
3. **Security:** Set SELinux to **permissive** and stopped the default firewall to allow mail traffic.

Set the hostname in the machine. **hostnamectl set-hostname nt67.edu**

```
File Edit View Search Terminal Help
[root@bashar ~]# hostnamectl set-hostname nt67.edu
[root@bashar ~]# vim /etc/host
host.conf hostname hosts
[root@bashar ~]# vim /etc/hosts
[root@bashar ~]# hostnamectl
    Static hostname: nt67.edu
            Icon name: computer-vm
            Chassis: vm
            Machine ID: 267c6cba842e44eea2e109f1bf5386aa
            Boot ID: a6c52b5c259f4ee4a4a9265ced743394
            Virtualization: microsoft
            Operating System: Red Hat Enterprise Linux 8.10 (Ootpa)
            CPE OS Name: cpe:/o:redhat:enterprise_linux:8::baseos
            Kernel: Linux 4.18.0-553.el8_10.x86_64
            Architecture: x86_64
[root@bashar ~]# vim /etc/hosts
```

Add hostname entries in /etc/hosts file as shown below-

vim /etc/hosts. Add the following **192.168.3.61 nt67.edu**

Disable or make permissive the SELINUX **setenforce 0** and check **getenforce**

Step 2: Installing and Configuring Postfix (SMTP)Postfix is responsible for sending emails.

- **Installation:** Installed using **dnf install postfix**.
- **Configuration:** Edited **/etc/postfix/main.cf** to define:

```
myhostname = nt67.edu
mydomain = nt67.edu
home_mailbox = Maildir/
```

- **Service:** Started and enabled the Postfix service.

Install EPEL Repository: ***dnf install https://dl.fedoraproject.org/pub/epel/epel-release-latest-8.noarch.rpm***

```
[root@bashar ~]# setenforce 0
[root@bashar ~]# getenforce
Permissive
[root@bashar ~]# dnf install https://dl.fedoraproject.org/pub/epel/epel-release-latest-8.noarch.rpm
Updating Subscription Management repositories.
Red Hat Enterprise Linux 8 for x86_64 - BaseOS 5.3 kB/s | 4.1 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - BaseOS 5.2 MB/s | 129 MB 00:24
Red Hat Enterprise Linux 8 for x86_64 - AppStream 5.9 kB/s | 4.5 kB 00:00
Red Hat Enterprise Linux 8 for x86_64 - AppStream 4.6 MB/s | 82 MB 00:18
Last metadata expiration check: 0:00:15 ago on Sat 09 May 2026 12:57:52 AM EDT.
epel-release-latest-8.noarch.rpm 14 kB/s | 25 kB 00:01
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Installing:
epel-release noarch 8-22.el8 @commandline 25 k
=====
```

3. Key Configuration Steps

- **Network Access:** Allowed Apache port **80** through the firewall.
- **Postfix Setup:** Configured the domain (nt67.edu) and set the mail storage format to **Maildir/**.
- **Dovecot Setup:** Enabled protocols (imap, pop3, lmtp) and configured plaintext authentication for local users.
- **Webmail Integration:** Configured **Apache (httpd)** with a Virtual Host to serve the SquirrelMail interface via **http://192.168.3.61/webmail**.

Allow the Apache default port 80 through your firewall/router:

firewall-cmd --permanent --add-port=80/tcp

Reload firewall using command: **firewall-cmd --reload**

Install the Postfix Package. **yum install postfix -y** and Configuring Postfix **vim /etc/postfix/main.cf**

```
81 history
[root@bashar ~]# yum install postfix -y
Updating Subscription Management repositories.
Extra Packages for Enterprise Linux 8 - x86_64 1.5 MB/s | 14 MB 00:09
Last metadata expiration check: 0:00:09 ago on Sat 09 May 2026 01:00:49 AM EDT
Package postfix-2:3.5.8-7.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@bashar ~]# vim /etc/postfix/main.cf
[root@bashar ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:03:1f:0b brd ff:ff:ff:ff:ff:ff
    inet 192.168.3.32/32 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
```

Implementation of Postfix and Dovecot Mail Server with Webmail Interface

Line no 94 - Uncomment and set your mail server FQDN *myhostname = nt67.edu* Line
102 - Uncomment and Set domain name = *mydomain = nt67.edu*

```
File Edit View Search Terminal Help
93 #
94 myhostname = nt67.edu
95 #myhostname = host.domain.tld
96 #myhostname = virtual.domain.tld
97
98 # The mydomain parameter specifies the local internet domain name.
99 # The default is to use $myhostname minus the first component.
100 # $mydomain is used as a default value for many other configuration
101 # parameters.
102 mydomain = nt67.edu
```

Line 118 - Uncomment - *myorigin = \$mydomain*
Line 132 - Uncomment and Set ipv4 - *inet_interfaces = all*
Line 138 - Change to all - *inet_protocols = all*

```
File Edit View Search Terminal Help
131 #
132 inet_interfaces= all
133 # Note: you need to stop/start Postfix when this parameter changes.
134 #
135 #inet_interfaces = all
136 #inet_interfaces = $myhostname
137 #inet_interfaces = $myhostname, localhost
138 inet_interfaces = all
139
140 # Enable IPv4, and IPv6 if supported
141 inet_protocols = all
```

Line 186 - *mydestination = \$myhostname, localhost.\$mydomain, localhost,*
Line 187 - Uncomment - *mydestination = \$myhostname, localhost.\$mydomain, localhost, \$mydomain*

```
File Edit View Search Terminal Help
185 #
186 mydestination = $myhostname, localhost.$mydomain, localhost
187 mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
188 #mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain,
189 # mail.$mydomain, www.$mydomain, ftp.$mydomain
```

Line 283 - Uncomment and add IP range - *mynetworks = 192.168.3.61/24, 127.0.0.0/8*

```
File Edit View Search Terminal Help
284 # (the value on the table right-hand side is not used).
285 #
286 mynetworks = 192.168.3.61/24, 127.0.0.0/8
287 #mynetworks = $config_directory/mynetworks
288 #mynetworks = localhost, localnet, localbroadcasts
```

Line 438 - Uncomment - *home_mailbox = Maildir/*


```
436 # mailbox file is /var/spool/mail
437 # "Maildir/" for gmail-style de
438 #
439 #home_mailbox = Mailbox
440 home_mailbox = Maildir/
```

Save and exit the file.

Start/restart Postfix service now:

```
systemctl restart postfix systemctl
```

```
status postfix
```

```
systemctl enable postfix
```

```
[root@zakaria ~]# systemctl status postfix.service
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; disabled; vendor pr>
   Active: active (running) since Wed 2026-05-06 02:40:53 EDT; 13s ago
     Process: 50601 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCE>
     Process: 50599 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, >
     Process: 50594 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, stat>
     Process: 50593 ExecStartPre=/usr/sbin/restorecon -R /var/spool/postfix/pid (c>
   Main PID: 50670 (master)
     Tasks: 3 (limit: 4140)
    Memory: 4.4M
    CGroup: /system.slice/postfix.service
            └─50670 /usr/libexec/postfix/master -w
              └─50671 pickup -l -t unix -u
                └─50672 qmgr -l -t unix -u
```

Test the Postfix Server:

Install Telnet Package **yum install -y telnet**

```
File Edit View Search Terminal Help
Last metadata expiration check: 0:03:28 ago on Sat 09 May 2026 01:18:17 AM EDT
No match for argument: telnet
Error: Unable to find a match: telnet
^[[A[root@bashar ~]# yum install -y telnet
Updating Subscription Management repositories.
Last metadata expiration check: 0:03:37 ago on Sat 09 May 2026 01:18:17 AM EDT
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
telnet x86_64 1:0.17-76.el8 rhel-8-for-x86_64-appstream-rpms 72 k
Transaction Summary
=====
Install 1 Package
Total download size: 72 k
Installed size: 119 k
Downloading Packages:
telnet-0.17-76.el8.x86_64.rpm 78 kB/s | 72 kB 00:00
-----
Total 78 kB/s | 72 kB 00:00
```

Check telnet - telnet localhost smtp Type the commands: **ehlo localhost** mail from:<bashar> rcpt to:<mail_server> must be a linux host in my network data **welcome to Postfix Mail Server**

quit

Implementation of Postfix and Dovecot Mail Server with Webmail Interface

```
mail from:<bashar>
250 2.1.0 Ok
rcpt to:<nt67>
250 2.1.5 Ok

data
354 End data with <CR><LF>.<CR><LF>
welcome to Postfix Mail Server
.
250 2.0.0 Ok: queued as 05BD060B3831
quit
221 2.0.0 Bye
Connection closed by foreign host.
[root@nt67 ~]# ls /home/nt67/Maildir/new/
1778304631.Vfd02I1000098M641552.nt67.edu
[root@nt67 ~]#
```

/home/mail_server/Maildir/new/ Open the file with name using cat. Install Dovecot Package **yum install -y dovecot**

```
[root@nt67 ~]# yum install -y dovecot
Updating Subscription Management repositories.
Last metadata expiration check: 2:02:58 ago on Sat 09 May 2026 01:18:17 AM EDT
Package dovecot-1:2.3.16-7.el8_10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
complete
```

Configure Dovecot **vim /etc/dovecot/dovecot.conf**

Line 24 - uncomment - *protocols = imap pop3 lmtp*

```
File Edit View Search Terminal Help
16 # Default values are shown for each setting, it's not recommended to
17 # change those. These are exceptions to this though: No section
18 # or plugin settings are added by default, they're listed here.
19 # Paths are also just examples with the real defaults being
20 # options. The paths listed here are for configuration --prefix
21 # --sysconfdir=/etc --localstatedir=/var
22
23 # Protocols we want to be serving.
24 protocols = imap pop3 lmtp submission
25
```

IMAP - Allow users to read email while leaving it on the server

POP3 - Allow users to download email to local devices LMTP

- Accept mail delivery directly from Postfix .

Edit file /etc/dovecot/conf.d/10-mail.conf file **vim /etc/dovecot/conf.d/10-mail.conf**

Line 24 - uncomment - *mail_location = maildir:~/Maildir*

```
File Edit View Search Terminal Help
22 # See doc/wiki/Variables.txt for full list. Some example
23 #
24 mail_location = maildir:~/Maildir
25 # mail_location = mbox:~/mail:INBOX=/var/mail/%u
26 # mail_location = mbox:/var/mail/%d/%ln/%n:INDEX=/var/
27 #
```

Edit /etc/dovecot/conf.d/10-auth.conf **vim /etc/dovecot/conf.d/10-auth.conf** And make the changes as shown below:

line 10 – uncomment- `disable_plaintext_auth = yes`

```
File Edit View Search Terminal Help
9 # See also ssl=required setting.
10 disable_plaintext_auth = yes
11
```

Line 100 - Add the word: "login" - `auth_mechanisms = plain login`

Edit file /etc/dovecot/conf.d/10-master.conf, **vim /etc/dovecot/conf.d/10master.conf** Make changes as shown below:

Line 102.,103 - Uncomment and add "postfix" -`mode = 0600 user = postfix group = postfix`

```
File Edit View Search Terminal Help
98 # something else than 0666 and Dovecot lets the kernel
99 # permissions (e.g. 0777 allows everyone full permissi
100 unix_listener auth-userdb {
101     #mode = 0666
102     user = postfix
103     group = postfix
104 }
```

Start Dovecot service: `systemctl`

`start dovecot`

`systemctl enable dovecot`

Testing Dovecot **telnet localhost pop3**

user bashar> pass> retr 1 > . > quit

```
[root@nt67 ~]# telnet localhost pop3
Trying ::1...
Connected to localhost.
Escape character is '^]'.
+OK Dovecot ready.
user bashar
+OK
pass
```

Install Squirrel mail Package `yum install squirrelmail`

sudo dnf install httpd php php-mbstring php-xml -y Step

2: Download and Extract SquirrelMail

```
wget https://sourceforge.net/projects/squirrelmail/files/stable/1.4.22/squirrelmail-webmail-1.4.22.tar.gz
tar -xvf squirrelmail-webmail-1.4.22.tar.gz sudo mv squirrelmail-webmail-1.4.22
```

```
/var/www/html/webmail
```

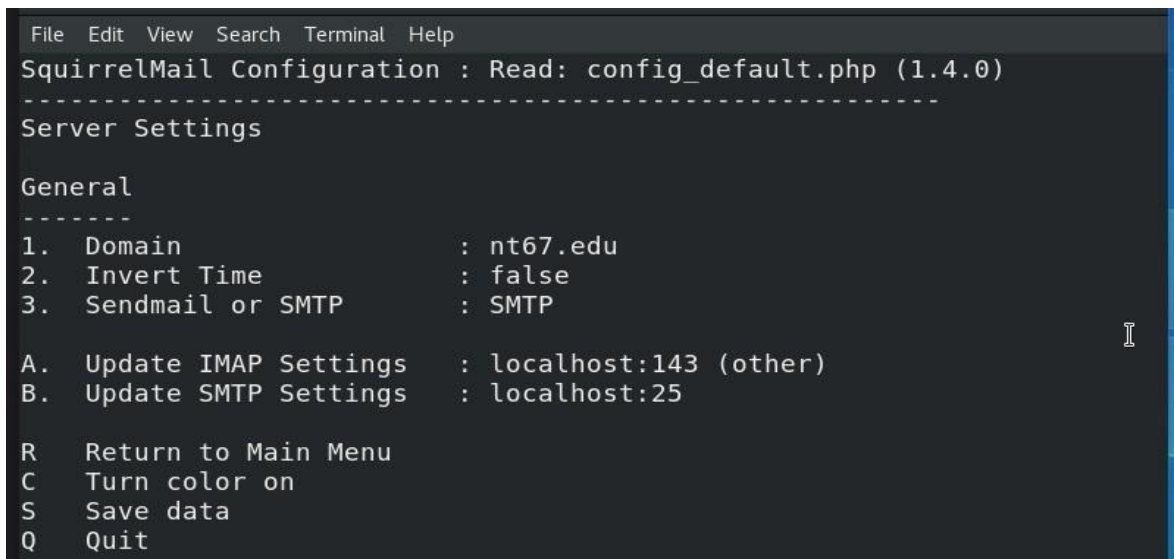
Set Permissions

sudo chown -R apache:apache /var/www/html/webmail edit

this file- **cd /var/www/html/squirrelmail/config/conf.pl**

1. Organization Preferences

1. Organization Name : SquirrelMail
2. Organization Logo : ../images/sm_logo.png
3. Org. Logo Width/Height : (308/111) 4. Organization Title : SquirrelMail \$version
5. Signout Page :
6. Top Frame : _top
7. Provider link : <http://squirrelmail.org/>
8. Provider name : SquirrelMail



```
File Edit View Search Terminal Help
SquirrelMail Configuration : Read: config_default.php (1.4.0)
-----
Server Settings

General
-----
1. Domain : nt67.edu
2. Invert Time : false
3. Sendmail or SMTP : SMTP

A. Update IMAP Settings : localhost:143 (other)
B. Update SMTP Settings : localhost:25

R Return to Main Menu
C Turn color on
S Save data
Q Quit
```

Exiting **conf.pl**.

You might want to test your configuration by browsing to <http://yoursqui/src/configtest.php>

Happy SquirrelMailing!

Create a squirrelmail vhost in apache config file: **vim /etc/httpd/conf/httpd.conf** Enter the following:

```
EnableSendfile on

# Supplemental configuration
#
# Load config files in the "/etc/httpd/conf.d" directory, if any.
IncludeOptional conf.d/*.conf
Alias /webmail /var/www/html/webmail
<Directory /var/www/html/webmail>
    Options Indexes FollowSymLinks
    RewriteEngine On
    AllowOverride All
    DirectoryIndex index.php
    Order allow,deny
    Allow from all
    Require all granted
</Directory>
```

systemctl restart httpd
systemctl enable httpd

```
[root@zakaria webmail]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor prese
   Drop-In: /etc/systemd/system/httpd.service.d
           └─php-fpm.conf
   Active: active (running) since Wed 2026-05-06 03:35:37 EDT; 3s ago
     Docs: man:httpd.service(8)
   Main PID: 58703 (httpd)
    Status: "Started, listening on: port 80"
     Tasks: 213 (limit: 4140)
    Memory: 33.0M
```

Enter the following command: **setsebool httpd_can_network_connect=1**

Verification & Testing

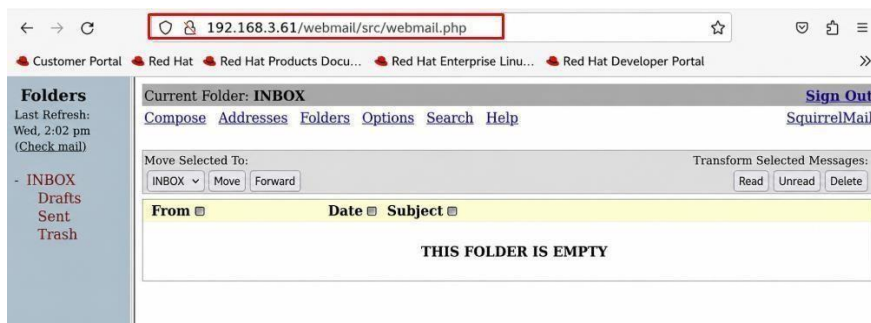
- **Postfix Test:** Verified sending mail via telnet on the SMTP port.
- **Dovecot Test:** Confirmed the mail server was ready to receive connections via the POP3 port.
- **Final Result:** Successfully logged into the **SquirrelMail** web interface and accessed the inbox using system user credentials.

Open Web Browser and type in address bar the following. And Use User Credentials to Login.

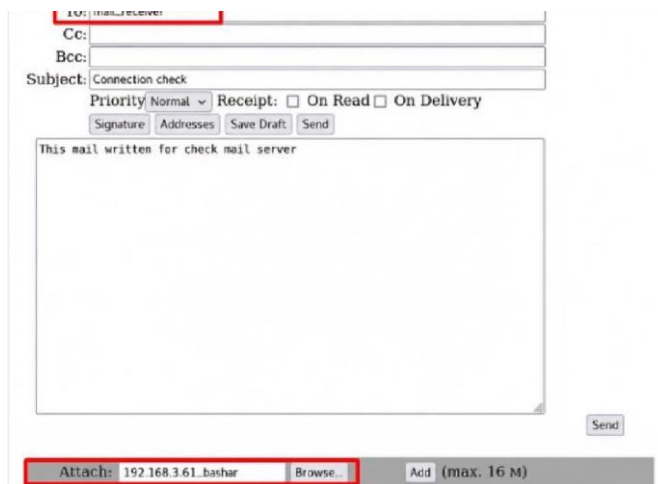
<http://192.168.3.61/webmail>



Your Dovecot Server is working



Final



Testing and Verification

- **Service Status:** Verified that all services (Postfix, Dovecot, Httpd) are active and running.
- **Web Access:** Accessed the mail client via <http://192.168.3.61/webmail>.
- **Final Result:** Successfully logged in with system credentials and confirmed that the Inbox is ready to receive messages.

Conclusion:

By implementing Postfix and Dovecot on RHEL 7, the company can achieve a secure, stable, and efficient internal mail server solution. Proper configuration of DNS, firewall, authentication, and encryption ensures reliable email communication and data protection.